Stem-cell decisions

Today I can… I can name a possible benefit and risk. I can ask for evidence that tests a claim. I can justify further research without claiming a cure.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Staff answers

Starter

1. Differentiation. 2. Two matching daughter body cells. 3. From a higher to a lower concentration overall. Confirm actual prior teaching rather than invent retrieval coverage.

I do

No. “It might help” is a possible benefit, not proof a treatment works. Evidence from an appropriate study is needed. The case and panel are fictional.

We do

Missing step: name an uncertain harmful result as a possible risk without asserting it has happened. Matching source order: evidence request; possible benefit; unsupported assertion; possible harm. “It must work because many people want it” is unsupported. A better question requests measured benefit and harm and how they were checked.

We do 2

Correct D. Check the possible benefit alongside missing risk evidence before deciding. A treats benefit as safety proof; B treats missing harm reporting as proof benefit did not occur; C generalises one study to everyone.

Hear BOTH learners independently. Proceed when both show the core idea. If either is unsure, re-model that step for that learner, then use the parallel hinge. The other learner can continue. No class-percentage rule is used.

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Staff answers · reasoning and limits

Parallel hinge

Another fictional study reports harms but no benefit measure. Can it establish a favourable benefit-risk balance? No. Both kinds of evidence are needed.

Independent

Supported: sort the prepared statements under benefit, risk and evidence needed; accept an accurate missing risk and evidence request. Independent: possible benefit alone does not establish effect or safety. Aspire: a qualified recommendation for further research must state a reason, risk and evidence request; distinguish research support from routine treatment approval.

Go deeper and research panel minute

Evidence of no improvement should reduce confidence in the claimed effect and change the recommendation, with reason. It is not proof about every possible study. Further research tests uncertainty under review; routine use requires stronger evidence of safety and benefit.

Misconceptions

A reported benefit in one study proves benefit for everyone: ask for further evidence across people and settings. A reported benefit establishes safety: consider missing harm data alongside benefit before deciding.

Fading

Full model: A fictional proposal may replace damaged cells. A possible harm is unknown. A careful study must measure both benefit and risk; possible use is not proven treatment.

Shared model leaves possible risk blank; learner identifies an uncertain harmful result. Independent has no filled answer. Next fade: leave the evidence-needed column blank so the learner chooses a discriminating test or observation.

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Staff guidance · pair access and evidence

Two-learner design

Use presenter and questioner roles, then switch. Each learner keeps an individual response; a shared product does not replace it. Offer private pointing, sign, drawing, writing, an exact scribe or staff-captured response. A learner may pass. No public cold calling or required personal disclosure. The case never asks for a real treatment decision or family medical history.

Adaptations

Use benefit/risk/evidence columns with one complete model sentence, then hide one cue. Give a structured partner reply and pass option. Check in privately, offer a quiet alternative and re-enter at the last completed step. Offer large text or accessible cards. Permit home-language planning and retain Independent/Aspire science demand.

Feedback and evidence

Use VF/WS/I/NS(+)/E/R/ // /? only for what occurred. Record each learner's hinge response, misconception, adult prompt and independent response separately. A real review edit earns //; read back on request ?. R follows a response to a next step heard and judged by an adult, never automatically. Capture E on [SYS] only for an observable curriculum action and never during distress. Use SCI-LAUNCH-A2-W03 with written, spoken, practical or photographic evidence.

If behind

Keep the core: decide what to test next when fictional safety results are missing. Preserve each learner's evidence. After a secure hinge shorten repeated rehearsal; if needed merge exit reflection into Week 4 Arrival and record reason and coverage impact. Pass on unresolved errors and next prompts. Do not substitute a group product.

Source and kit limits

Source curriculum mapping is masked [AWARD]/[QUAL]. Prepared model pieces are optional rehearsal, not a core practical or real cell study. Staff confirm school risk assessment and exact kit. Cards stay on the mat; pick up any dropped cards.

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Staff guidance · evidence and invitations

Source curriculum and evidence record

Foundation mapping: [AWARD] [QUAL] Biology Paper 1 and Combined Science Biology Paper 1, shared point 2.9: medical stem-cell benefits and risks. Consent is a research-ethics extension, not an extra specification point. No Biology-only or higher-tier point is assumed for Combined Foundation. The workbook approach does not create a new specification point.

Scheme evidence: [AWARD] [QUAL] Paper 1 foundation topic tests; [AWARD] [QUAL] science units. Formative: topic quizzes and practical write-ups. Summative: Topics 1 to 3 assessment in December. Here retain each individual explanation, review edit and exit beside the original response on [SYS].

Retrieval bindings: differentiation from Autumn 2 Week 2; mitosis and cell splitting from Autumn 2 Week 1; diffusion from Autumn 1 Week 4. “Last term” means the workbook Autumn 1 block. Check actual teaching if a literal previous academic term is required.

Careers link: a research ethics coordinator considers participant safety, evidence and consent before a clinical study.

Optional invitation bank · learner may pass

Route / invitation	Expected answer
Supported: What word means a possible good result?	Benefit.
Supported: What word means information used to test a claim?	Evidence.
Independent: Why is “might help” not proof of a treatment?	It states a possibility; collect evidence of benefit and risk.
Independent: Name one information request for the fictional study.	Ask for measured benefit or safety outcomes and how they were checked.
Aspire: How could new evidence change a panel decision?	It can raise or lower confidence in benefit and risk, changing support for further work.
Aspire: Why support research without approving routine treatment?	Research tests uncertainty under review; routine use needs stronger safety and benefit evidence.

Individual targets and evaluation

Cognition: record the missing-risk task and sequencing, attention or independence with prompt level. Communication: record the hinge in the agreed mode without requiring speech. Regulation: record the private Arrival or pause/re-entry target. Access: record accessible Independent input and target-related independence, not diagnosis. Language: record first-language planning and oral rehearsal with the four actual words; EAL is not SEN.

Capture E against the learner’s own individual plan target on [SYS], including linked education, health, care and personal targets. Follow the learner’s individual support/behaviour plan where it differs. Staff record date taught; delivered as planned; criteria met; adaptations and reasons; coverage impact; and evidence banked. No pupil data is stored in HTML.

Made from the web lesson SCI-LAUNCH-A2-W03 · version _____

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